

Computational Geometry

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September 2024

1 Introduction

Computational Geometry is the study of data structures and algorithms for geometric objects. A good algorithmic solution to the geometric problems require the understanding of the geometric properties of the given problem as well as effective algorithmic techniques that can be applied to forge solutions that are asymptotically fast. This report will focus only on the geometric problems, techniques for solution and properties of these solutions that is obtained.

2 Preliminaries

2.1 Set partion

Suppose there is a set of P points, $P := \{P_1, P_2, \dots, P_n\}$. Then the set partitioning of these points mean to divide each points into disjoint sets. Mathematically partitioning means,

1. $P = \{P_1\} \cup \{P_2\} \cup \{P_2\} \cup \dots \cup \{P_n\}$
2. If $i \neq j \Rightarrow P_i \cap P_j = \emptyset$

2.2 Half plane

A planner region containing all points on one side of the infinite straight line and none from the other side is called a half plane. For example, a plane xy, is divided into two half planes by the line, $y = 2x$. One half plane contains all the points for which the inequality holds $y > 2x$ and the other half plane contains all the points for which the inequality holds that $y < 2x$

2.3 Graphs

A graph can be defined as $G = (V, E)$, where V is the non-empty set of vertices and E is possibly empty set of edges. If an edge say, $\{u, v\} \in E$, then vertex u and vertex v are connected. Two vertices are adjacent if $\{u, v\} \in E$. An edge is incident on vertex u and vertex v. Vertices adjacent to vertex u are neighbours

of u , denoted as $N(u)$. The number of neighbours of vertex v is called the degree of v , denoted as $deg(v) = |N(v)|$. If there are differences in neighbours among vertices, then the minimum degree and maximum degree is denoted as $\delta(G)$ and $\Delta(G)$ respectively.

A graph is called a planar graph if it can be drawn without any edge crossing. The definition says "if it can be drawn" as some graphs which at first glance don't seem to be planar, but can be redrawn in a way that their edges don't cross. The planar graph divides the plane in different regions which are called faces. The number of faces is constant and it doesn't matter how a planar graph is redrawn, as long as the defining constraints are not violated, the number of faces is going to be the same. One of the faces that is always present in the graph is the unbounded face.

For planar graph the Euler's formula states,

$$V - E + F = 2$$

Here V, E and F are number of vertices, edges and faces respectively.

2.4 Convex Hull

Convex Set: $S \subset \mathbb{R}^2$, is called convex, iff $\forall p, q \in S$, the line segment $\overline{pq}$ is completely contained in S .

A convex hull of S , denoted as $\text{conv}(S)$, is the smallest convex set containing S . Convex hull of S , is a unique convex polygon whose vertices are points from S and in the polygon all the points belonging to S is contained. The word unique is used to specify that, given the same set of vertices the same(unique) convex polygon will be obtained.

If there is only one point convex hull is a point. If there are two points convex hull is a line. If there are 3 points convex hull is the triangle.

For example, say there is a point set, $S := P_1, P_2, \dots, P_9$ and each point P_i 's positioning visualized in figure 1. Then the convex hull of S is defined as the smallest convex set containing all points of S , which is, $\text{conv}(S) = \{P_2, P_4, P_5, P_8, P_9\}$

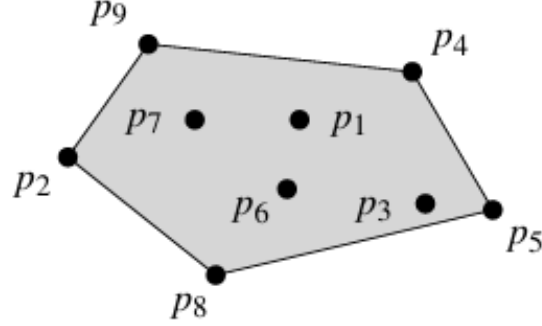


Figure 1: Convex Hull of S. Image credit, reference (1)

3 Voronoi Diagram

3.1 Background

Voronoi diagrams are techniques that are used partition sets, where set $P \subset \mathbb{R}^n, n \geq 2$. Though the voronoi diagram can be generalized for sets that belong to higher dimensional spaces, the report only contains discussion abouts sets that belong in 2D.

Lejeune Dirichlet and M.G Voroni are credited for the formalizaiton of the set partitioning technique. For this reason this technique is also known as Dirichlet's tesellation.

For the formal definition some terminologies need to be introduced. Each point on a set is a site and the disjoint regions these points are going to be divided into are called cells.

3.2 Definition

Say a given set S with the points $P_1, P_2, P_3, \dots, P_n$ is on the plane, Voronoi diagram divides the plane into n voronoi regions where each region satisfies the following properties.

1. Each P_i lies in one voronoi region
2. If an arbitrary point in plane q ($q \neq S$) is inside a voronoi region with site P_i , then

$$\sqrt{(q_x - P_{ix})^2 + (q_y - P_{iy})^2} < \sqrt{(q_x - P_{jx})^2 + (q_y - P_{jy})^2}$$

where $j \neq i$ and $1 \leq i, j \leq n$

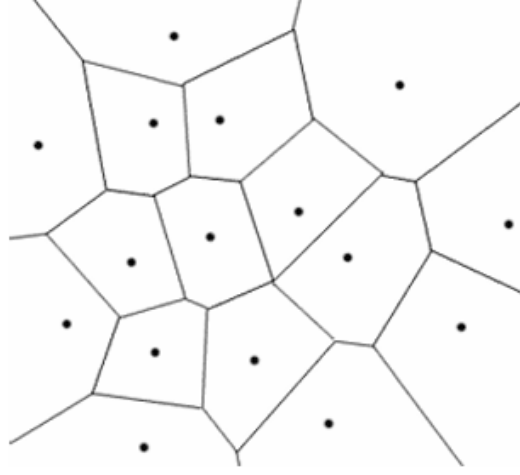


Figure 2: A Voronoi Diagram. Image credit, reference(6)

More simply, an arbitrary point in the 2D plane, belongs to the cell with site P_i , if the Euclidean distance from the arbitrary point and the site is smaller than the Euclidean distance from the the arbitrary point to any other site.

3.3 Voronoi regions

The Voronoi diagram consists of different Voronoi regions also known as cells. The line segments that form the boundaries of the Voronoi regions are called Voronoi edges. A Voronoi edge is simply the perpendicular bisector of the connecting line segments in between two Voronoi sites. To understand the edges better the Voronoi diagram of a set containing only two points can be considered. The Voronoi edge for this case is an unbounded line that bisects the line segment connecting the two points. This infinite line forms two half planes. For more than two points, the voronoi edge is simply an extension of the case described above.

As established above, that the Voronoi edges divide the entire plane into half-planes, the Voronoi region definition can be formalized from there. Voronoi region $Vor(P_i)$ is defined as the intersection of $n - 1$ half planes formed by taking the perpendicular bisector of the line segment $\overline{P_i P_j}, \forall P_j \in S$ where $i \neq j$ and $\forall j, P_i P_j$ are neighbors sharing a common boundary. This means,

$$Vor(P_i) = \bigcap_{j \neq i} H(P_i P_j)$$

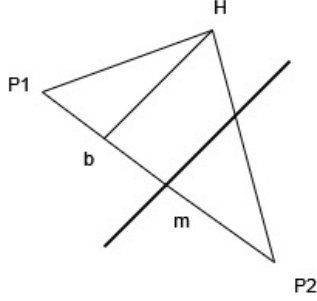


Figure 3: For proof of theorem 1

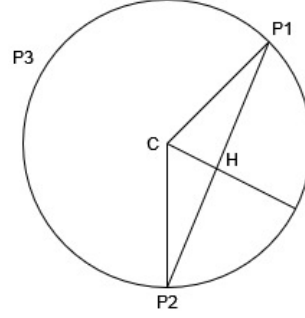


Figure 4: For proof of theorem 2

The entire Voronoi region can be computed as the following,

$$Vor(S) = \bigcup_{i \in S} Vor(P_i)$$

It is to be noted that the Voronoi regions can be bounded or unbounded. The regions are unbounded when the Voronoi sites are at the extremes (see figure 2). Additionally, the end points of the intersecting edges are called the Voronoi vertices.

3.4 Properties of the Voronoi diagrams

1. Number of Voronoi vertices is at most $2n - 5$
2. Number of Voronoi edges is at most $3n - 6$
3. Each Voronoi vertex is an intersection of exactly 3 points
4. If site $P_i \in S$ is a neighbor of $P_j \in S$ then the Voronoi region $Vor(P_i)$ and $Vor(P_j)$ will have a common edge
5. $Vor(P_i)$ is unbounded iff P_i is an extreme point of S . (See figure 2)
6. If $Vor(a)$, $Vor(b)$ and $Vor(c)$ are neighbouring Voronoi regions with three neighboring sites a, b, c , then a triangle $\triangle abc$ can be formed by connecting these sites and the center of the circumcircle that goes through these sites, is a Voronoi vertex.

3.5 Proofs

3.5.1 Theorem 1

All points on the half-plane containing $P_1 \in S$ and delimited by the perpendicular bisector L of $\overline{P_1P_2}$, are closer to P_1 than P_2

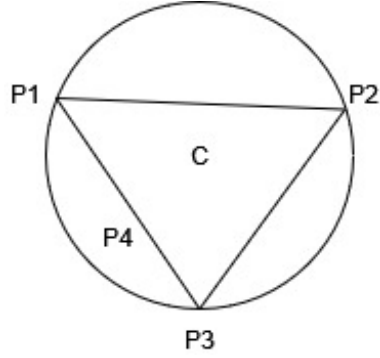


Figure 5: For proof of theorem 3

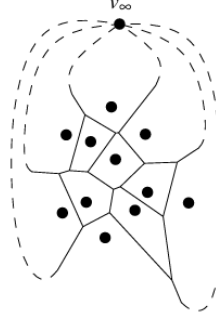


Figure 6: For proof of theorem 4.
Image credit, reference (1)

Proof:

Say, H is an arbitrary point in the plain(see figure 3). Applying Pythagorean Theorem,

$$(P_1H)^2 = (P_1b)^2 + (Hb)^2$$

$$(P_2H)^2 = (P_2b)^2 + (Hb)^2$$

$$\text{Since } P_2b > P_1b \Rightarrow (P_1H)^2 < (P_2H)^2$$

3.5.2 Theorem 2

The intersection of the 3 perpendicular bisectors P_1, P_2, P_3 is the center of the circle containing them

Proof:

Say C is the center of the circle. The line perpendicular to $\overline{P_1P_2}$ that intersects at C is $\overline{CH}$ (see figure 4). Now this is sufficient to prove that for $\triangle P_1P_2C$, $\overline{CH}$ is the perpendicular bisector of $\overline{P_1P_2}$ and this argument for $\triangle P_1P_2C$ holds for the other triangles $\triangle P_2P_3C$ and $\triangle P_1P_3C$. So, for $\triangle P_1CH$ and $\triangle P_2CH$,

$$P_1C = P_2C, \overline{CH} \text{ common and } \angle P_1HC = \angle P_2HC = 90 \text{ deg}$$

This means the triangles are congruent, meaning $\overline{P_1H} = \overline{P_2H}$. This proves that the line $\overline{CP}$ is a perpendicular bisector of $\overline{P_1P_2}$. The same argument can be extended for the other triangles.

3.5.3 Theorem 3

The circle containing the voronoi vertex v and passing through 3 points $P_1, P_2, P_3 \in S$ is empty

Proof:

The circle is empty means there are no other point $P_i \in S$, in the interior of the

circle. A proof by contradiction can be constructed to prove the theorem.

Say, the negation of the theorem 3 is true, which means the containing circle is non-empty(See figure 5). This implies there is a point say, P_4 inside the circle. Since P_4 is inside the circle, the center C should be the closest to P_4 and this means C should be contained in the Voronoi region $Vor(P_4)$. But in actuality, C is contained in all of $VOR(P_1), VOR(P_2), VOR(P_3)$. Therefore, the negation of the theorem 3 is proven to be false and the theorem 3 is true.

3.5.4 Theorem 4

For $n \geq 3$, the number of vertices in the Voronoi diagram of a point set of n points/sites, is at most $2n - 5$ and number of edges atmost $3n - 6$.

Proof: For this proof the Euler's graph formula is required. But since the Voronoi diagram contains half infinite edges(defined on one end but undefined on the other), it is not a planner graph. So a vertex at infinity is taken and connected to the unbound edges, to make it a proper connected planner graph (see figure 6). Now, every vertex in Voronoi has degree 3, but V_∞ can have more. For graph it is known that the sum of degree in a graph is equal twice the number of edges. So for this situation, it can be written,

$$2E \geq 3(V + 1)$$

where E = No. of edges, V = No. of vertices. It is also known from the Euler's formula that,

$$V - E + F = 2$$

where E = No. of edges, V = No. of vertices, F = number of faces. The number of faces in Voronoi diagram is equal the number of sites = n . Now from the above discussion the following inequalities can be constructed,

$$\begin{aligned} E &\geq 3(V + 1)/2 \Rightarrow -E \leq -3(V + 1)/2 \\ \Rightarrow (V + 1) - E + n &\leq (V + 1) - 3(V + 1)/2 + n \\ \Rightarrow 2 &\leq -(V + 1)/2 + n \\ \Rightarrow 4 - 2n &\leq -(V + 1) \\ \Rightarrow 2n - 5 &\geq V \\ \Rightarrow V &\leq 2n - 5 \end{aligned}$$

In the same manner, it can be shown that $E \leq 3n - 6$

3.6 Voronoi diagram to Delaunay triangulation

Delaunay triangulation is the dual of Voronoi diagram. It is possible to obtain delaunay triangulation from a Voronoi diagram. To obtain the triangulation, the Voronoi sites are connected by an arc iff $Vor(P_i)$ and $Vor(P_j)$ are bounded by a common Voronoi edge. This basically creates a planar graph connecting all the vertices in set S across the edges of the Voronoi regions, which is nothing but the Delaunay triangulation.

4 Delaunay Triangulation

4.1 Background:

Delaunay triangulation is the dual of Voronoi diagram. It is a technique or algorithm, to triangulate a given point set. The triangulation can be done at random but what separates those triangulations from Delaunay triangulation is that the angles of the triangles are maximized as well as some other properties relating to interpolation is optimized in Delaunay triangulation. Useful applications of the Delaunay triangulations are terrain approximation, mesh generation, surface approximation etc. (see figure 8)

The result from the Delaunay triangulation is a geometric structure sometimes called simplicial complex. This simplicial complex is made up of simplices(triangles).

Delaunay triangulation comes from Delaunay graphs. Delaunay graphs or triangulation is named after the Russian mathematician Boris Nikolaevich Delone. Since his work was published in French his name is better known in the French transliteration.

To move towards the definition of the Delaunay triangulation firstly triangulation needs to be formally defined.

4.2 Triangulation

A triangulation of finite point set $P \subset \mathbb{R}^2$ is a collection of triangles τ such that,

1. Convex hull of $P = \bigcup_{T \in \tau} T$
2. Discrete point set $P = \bigcup_{T \in \tau} \text{Vertices}(T)$
3. For every distinct pair of triangles say T and $U \in \tau$, the intersection, $T \cap U$ is either common vertex, common edge or empty set($\emptyset$)

Triangulation of point set, polygon, polyhedron etc are possible, but this report will only focus on point set.

4.3 Angle vector

Let τ be triangulation of p and it has m triangles. If all the angles are sorted by an increasing value ($\alpha_i \leq \alpha_j, \forall i < j$) and the resulting sequence is: $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{3m}$, then $A(\tau) = \langle \alpha_1, \alpha_2, \alpha_3, \dots, \alpha_{3m} \rangle$ is called the angle vector. Usually, for a given point sets there exists multiple triangulations and say $A(\tau')$ is the angle vector of another triangulation. Now,

If $\{\exists i | 1 \leq i \leq 3m\}$ for which $\alpha_j = \alpha'_j, \forall j < i$ and $\alpha_i > \alpha'_i \Rightarrow A(\tau) \geq A(\tau')$

If $A(\tau) \geq A(\tau'), \forall \tau'$ of $P \Rightarrow A(\tau)$ is the optimal triangulation

4.4 Illegal edge and edge flip

Say P_i, P_j, P_k and P_l form a quadrilateral (see figure 9). Of this quadrilateral, if the edge $\overline{P_i P_j}$ is considered then $\triangle P_i P_j P_k$ and $\triangle P_i P_j P_l$ are formed. If the edge $\overline{P_l P_k}$ is considered then $\triangle P_l P_k P_i$ and $\triangle P_l P_k P_j$ are formed. Now, say $A(\tau)$ is the angle vector for triangulation that is obtained when taking edge $\overline{P_i P_j}$ and $A(\tau')$ is the angle vector for triangulation that is obtained when taking edge $\overline{P_l P_k}$. The edge $e = \overline{P_i P_j}$ is illegal if,

$$\min \alpha_i < \min \alpha'_i \text{ where } 1 \leq i \leq 6$$

Simply an edge is illegal if the minimum angle can be extended. When the edge is rendered illegal it can be flipped (the other edge is considered) to obtain the optimal triangulation. Another way to prove if the edge is illegal or not is through a lemma that states, if P_i, P_j, P_k, P_l don't reside on the same circle then there is exactly one illegal edge and the illegal edge is the one for which one of the points of the quadrilateral falls within the circumcircle. If all four points are cocircular then both the edges are legal. A triangulation which doesn't contain any illegal edges, is called a legal triangulation.

4.5 Delaunay definition

A triangulation of a finite point set $P \subset \mathbf{R}^2$ is called a Delaunay triangulation, if the circumcircle of every triangle is empty, that is, there is no point from P in its interior. From another point of view, any triangulation that is legal is Delaunay triangulation. In the later sections this statement will be proven.

4.6 Properties of the Delaunay triangulation

1. No two edges intersect with each other
2. All circumcircle is empty
3. Two sites P_i and P_j are connected by an edge iff there is an empty circle passing through P_i and P_j
4. The closest pair of sites are neighbours in the Delaunay triangulation

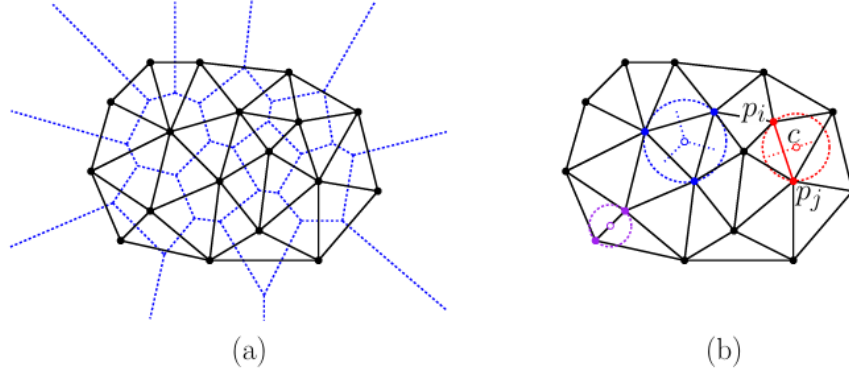


Figure 7: (a) Voronoi edges in blue and Delaunay triangulation in black (b) Delaunay triangulation and circumcircles. Image credit, reference(12)

5. All the properties of triangulation stated above also hold

4.7 Proofs

4.7.1 Intersection property

No two edges in the Delaunay triangulation, τ intersect with each other

proof

This directly comes from section 3.6, where it is described that delaunay triangulation forms a planner graph. In planner graphs the edges cannot intersect with each other

4.7.2 Circumcircle property

The circumcircle that goes through the vertices of any Delaunay triangle is empty, that is no other sites apart from the vertices are contained in the circumcircle.

Proof:

This property is proven with the help of the voronoi diagrams. In voronoi diagram the voronoi vertex is the intersection of the perpendicular bisectors of exactly three edges that connect the nearest neighbors. Since the three nearest neighbors from the voronoi vertex(center of the circumcircle) no other points are close enough to be contained inside the circumcircle.

4.7.3 Empty circle property

There exists an edge between P_i and P_j in the Delaunay triangulation τ , iff there is an empty circle passing through P_i and P_j

Proof:

The proof directly follows from the circumcircle property. Since by definition

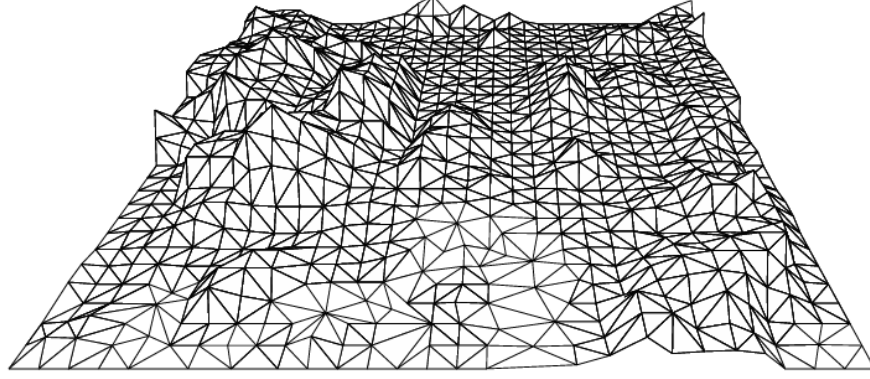


Figure 8: Terrain approximation using Delaunay triangulation. Image credit, reference (1)

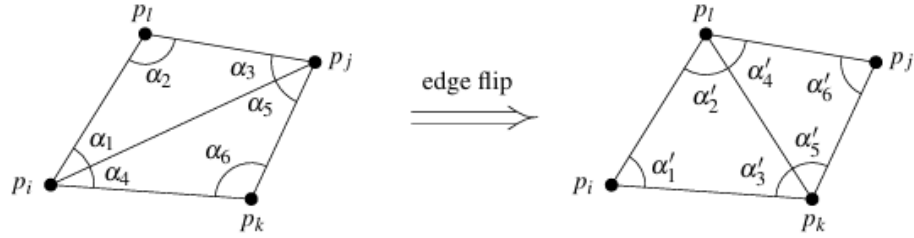


Figure 9: Edge flip illustration. Image credit, reference (1)

the circumcircle through the Delaunay triangles vertices is empty, it can also be stated without any loss of validity, that there exists an empty circle that passes through the two sites that form an edge in Delaunay triangle.

4.7.4 Nearest pair property

The closest pair of sites are neighbors(have an edge) in Delaunay triangulation.

Proof

Say the closest pair are P_i and P_j . Since they are taken to be the closest pair of sites, to prove they are neighbors it is sufficient to prove the edge $\overline{P_i P_j}$ exists. Now, an edge exists if an empty circle goes through P_i and P_j . If an empty circle is drawn keeping the two edges in the two ends of the diameter, no other point is contained in that circle. Since an empty circle can be drawn, there exists an edge, which means P_i and P_j are neighbors. See figure 7(b) purple circle at the bottom left.

4.7.5 Theorem 5

Let P be a point set of n points. Say there are k points on the convex hull of P . Then triangulation will give exactly $2n - 2 - k$ triangles and $3n - 3 - k$ edges.

Proof

In triangulation τ of the point set P with n points, let m be the number of triangles. The number of faces then is $F = m + 1$ where 1 is added because of the unbounded face. Now, each triangle has 3 edges in the bounded region and k edges in the unbounded region (because k edges make up the convex hull), so the number of edges $E' = 3m + k$. But as it is also known that each edge is common for two faces which means each edge is counted twice. So the actual number of edge is actually, $E = (3m + k)/2$

Now according to Euler's formula, defined in section 2.3,

$$\begin{aligned} V - E + F &= 2 \Rightarrow n - (3m + k)/2 + m + 1 = 2 \\ \Rightarrow n - 3m/2 - k/2 + m &= 1 \Rightarrow -m/2 = -n + k/2 + 1 \\ \Rightarrow m &= 2n - k - 2 \end{aligned}$$

In the similar manner, if the value of m is substituted as $m = (2E - k)/3$ in the Euler's formula it can be proven that $E = 3n - k - 3$.

4.7.6 Corollary

Triangulation of a set $P \subset \mathbb{R}^2$ of $n \geq 3$ points τ is maximum planner iff the convex hull is a triangle.

Proof:

This proof can directly be established from theorem 4 and 5. A maximum planner graph is such that, it has maximum number of legal edges. From theorem 4 it is known that there can be at most $3n - 6$ edges are possible in the voronoi diagram. So a maximum planner triangulation would mean that there will be $3n - 6$ edges. From theorem 5 it is known that, the number of edges in Delaunay triangulation is $3n - k - 3$. Now, the number of edges in Delaunay triangulation is equal to the maximum number of edges iff $k = 3$ and $k = 3$ means there are three points on the convex hull of P . This implies the convex hull is a triangle.

4.7.7 Theorem 6

Let a point set $P \subset \mathbb{R}^2$ of n points be on a plane. Then a triangulation τ of P is legal iff τ is Delaunay.

Proof:

The proof can be done through contradiction. Assume, there is a triangulation that is legal but not Delaunay.

Say, legal triangulation of P is τ . Since τ is legal but not Delaunay, this means τ can have non-empty circumcircles. But following from the lemma of legal

triangulation, stated in section 4.4, the legal triangulation cannot have any sites(vertices) inside the circumcircle. In other words the circumcircle of the legal triangles must be empty. Since the initial assumption is not true, the negation of the assumption is true.

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